

Software Engineer UI Developer

Take-Home Exercise: Login + MFA

Objective

Build a simple authentication experience demonstrating login, MFA, form validation, and basic access control. We want to evaluate your technical skills, design decisions, and problem-solving approach.

Requirements

1. Login & MFA

- Login screen with email and password.
- Validate email and password fields with clear error messages.
- Add a second authentication step using a mock MFA method such as OTP, email code, or authenticator code.
- Include a simple Sign Up flow that navigates to a separate screen. Full registration is not required.
- Flow: **Login** → **MFA** → **Protected Screen**

2. Access Control

- After successful login and MFA, redirect to a protected screen.
- Demonstrate two user roles:
 - **Read-only:** Edit actions are hidden or disabled.
 - **Read/write:** Edit actions are visible and enabled.
- Mock users/roles are acceptable; no backend authentication is required.

4. Technical

- Use React
- Keep the code modular and maintainable.
- Include tests where appropriate.
- UI/component libraries are allowed.

AI Usage

You may use AI tools such as ChatGPT, Claude, GitHub Copilot, etc.

You should be able to explain:

- The code you submit.
- Your technical decisions and trade-offs.
- Any AI-generated code you used.

Submission

Please provide:

- **GitHub repository** with the completed exercise.
- **README** including:
 - Technologies used
 - Setup/install instructions
 - Local run instructions
 - Mock user credentials/roles
 - How to test the login/MFA flow
 - Key design decisions and assumptions
 - Known limitations
- **Demo** showing:
 - Login and MFA flow
 - Read-only and read/write experiences
 - Validation/error handling
 - How to run the application locally

Evaluation

We will evaluate correctness, code quality, application structure, access-control implementation, UI/UX, testing approach, and your ability to explain your implementation and trade-offs.

There is no single "correct" solution. We are interested in how you approach the problem and the decisions you make.